

Neural Population Dynamics Reveal Internal Representations Underlying Adaptive Obstacle Avoidance in Reinforcement Learning

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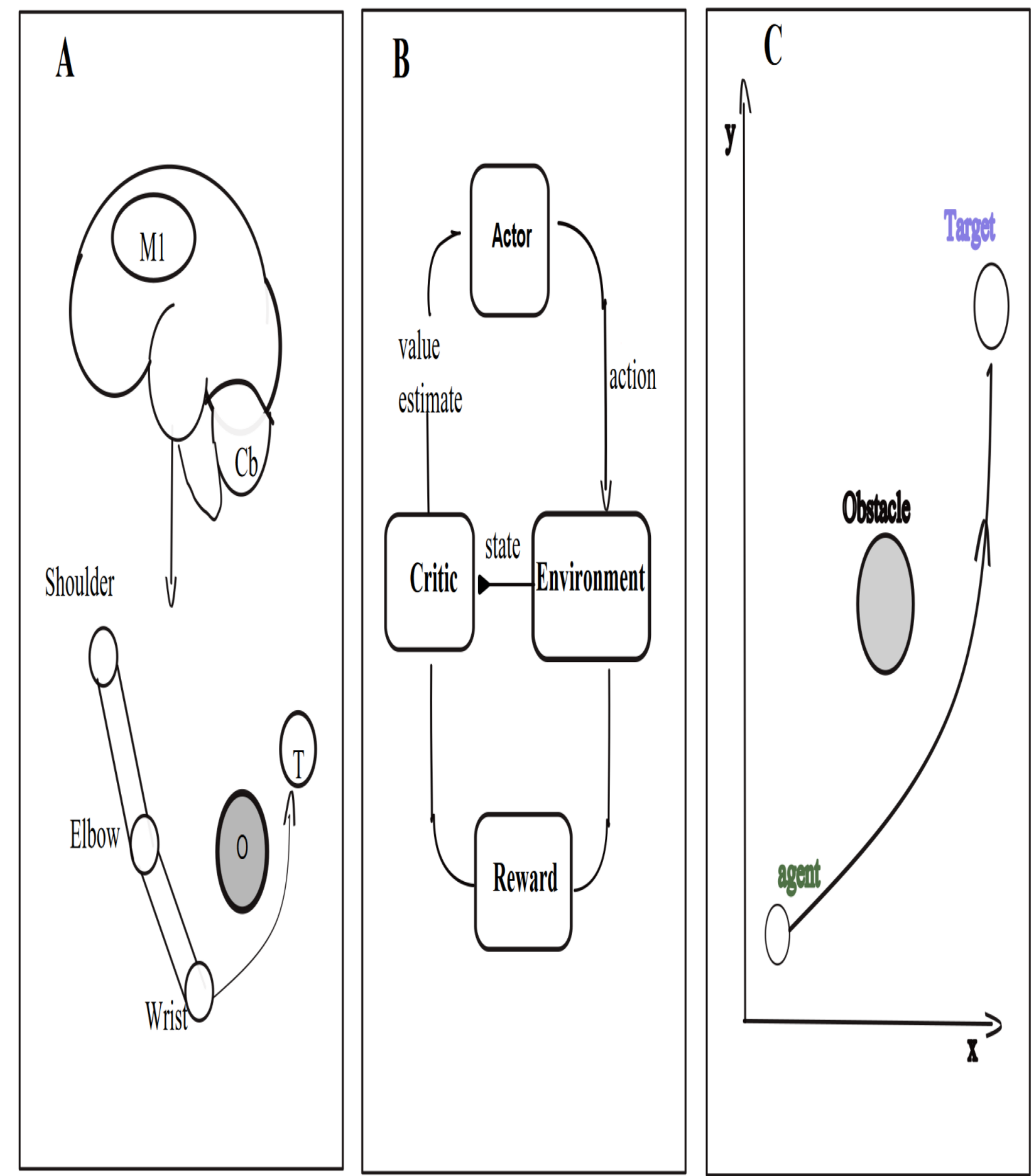
BACKGROUND & MOTIVATION

- Motor recovery after stroke is highly heterogeneous. Existing therapies lack mechanistic precision to tailor training to the individual patient.
 - Reinforcement learning (RL) controllers trained by reward discover adaptive motor strategies analogously to biological motor learning.
 - Motor cortex population activity evolves on compact low-dimensional manifolds. Whether RL controllers develop the same organisation is unknown.
- We ask: does reward-driven learning produce structured neural population representations parallel to biological motor cortex?

METHODS

- 2-D point-mass agent navigates obstacle corridor to a fixed goal under Newtonian dynamics (Gymnasium environment).
- PPO actor-critic controller trained for 3×10^6 steps; two hidden layers $\times$ 256 units (feedforward MLP).
- 7 perturbation conditions: control (P0), leftward (L1–L3) and rightward (R1–R3) lateral force impulses.
- Population analysis: PCA · K-means · linear decoding applied to hidden-layer activations during evaluation.

Fig 1 · Task & Control Architecture



KEY FINDINGS

- 74.8 % variance in 2 PCs**

Hidden-layer activity compresses into a compact low-dimensional manifold — mirroring motor cortex.
- 4 phase-aligned clusters**

K-means reveals movement-phase structure (approach · perturbation · correction · arrival).
- Direction asymmetry**

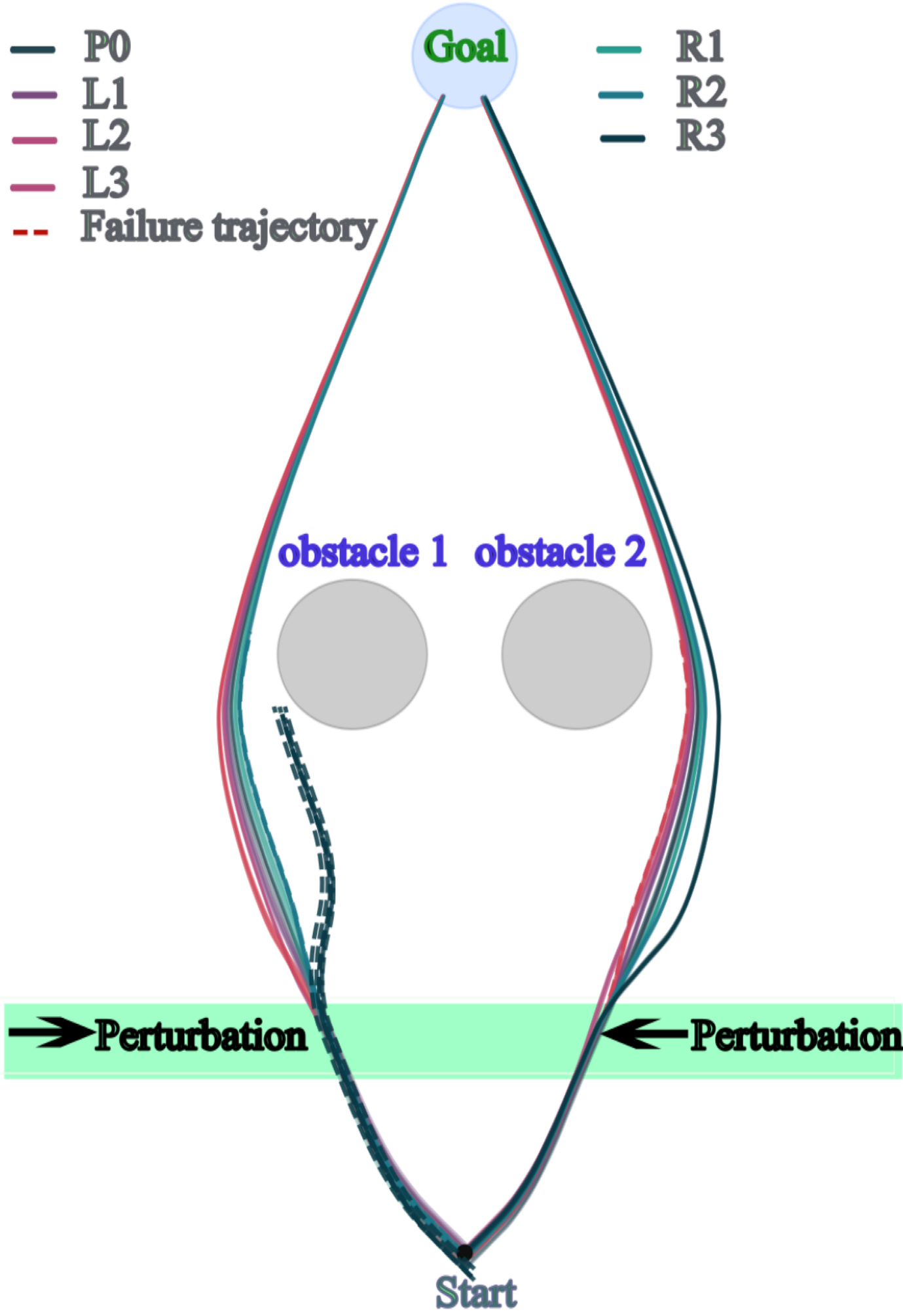
Rightward perturbations are more disruptive due to learned left-biased corridor trajectory.
- Linear decodability**

Extreme conditions linearly separable; mild conditions geometrically indistinct — matching biological findings.

RESULTS

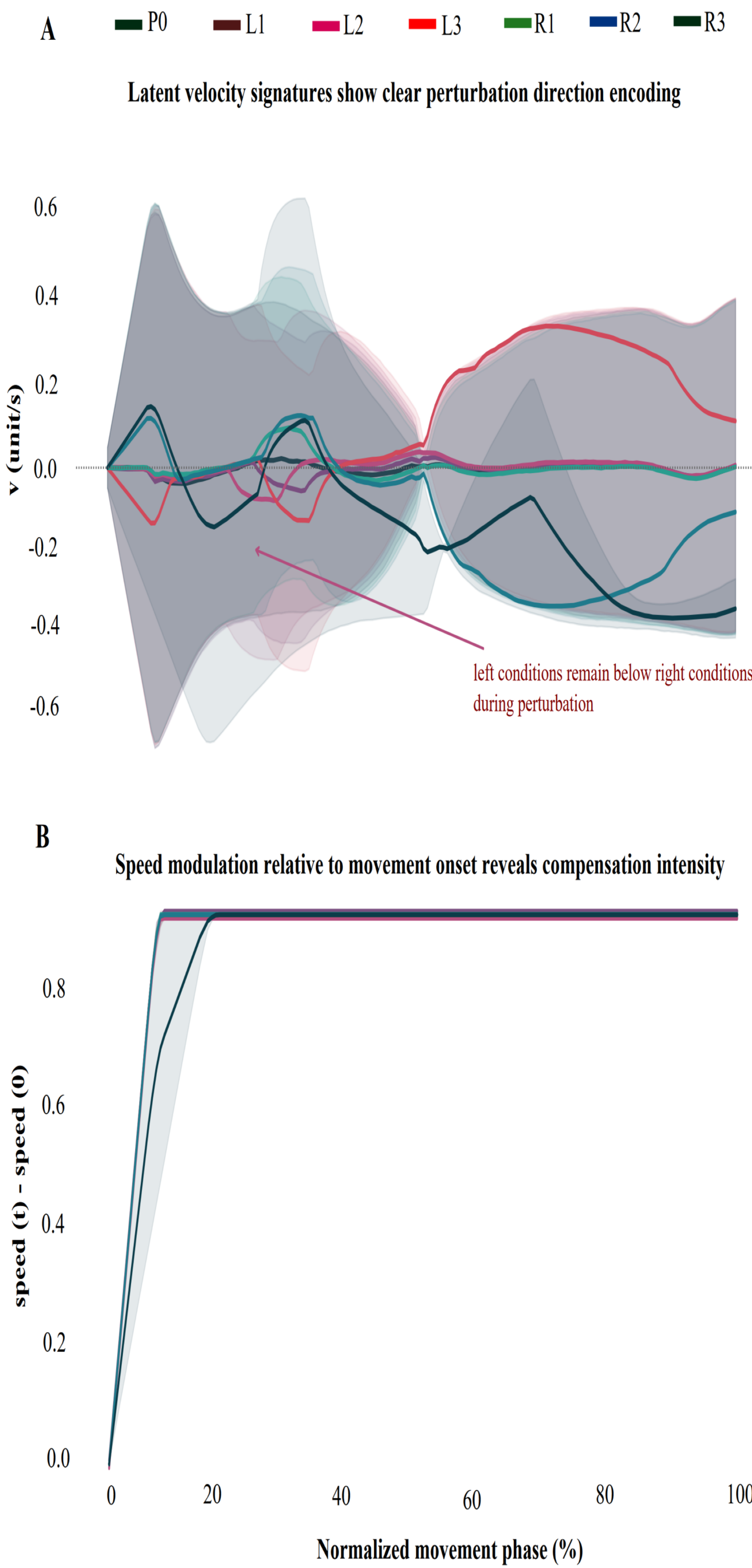
Fig 2 · Adaptive Route Geometry Under Lateral Perturbation

Adaptive route geometry under graded lateral perturbation



Solid = success · Dashed = failure · Shaded band = perturbation zone · Rightward conditions (R2, R3) show highest failure rate

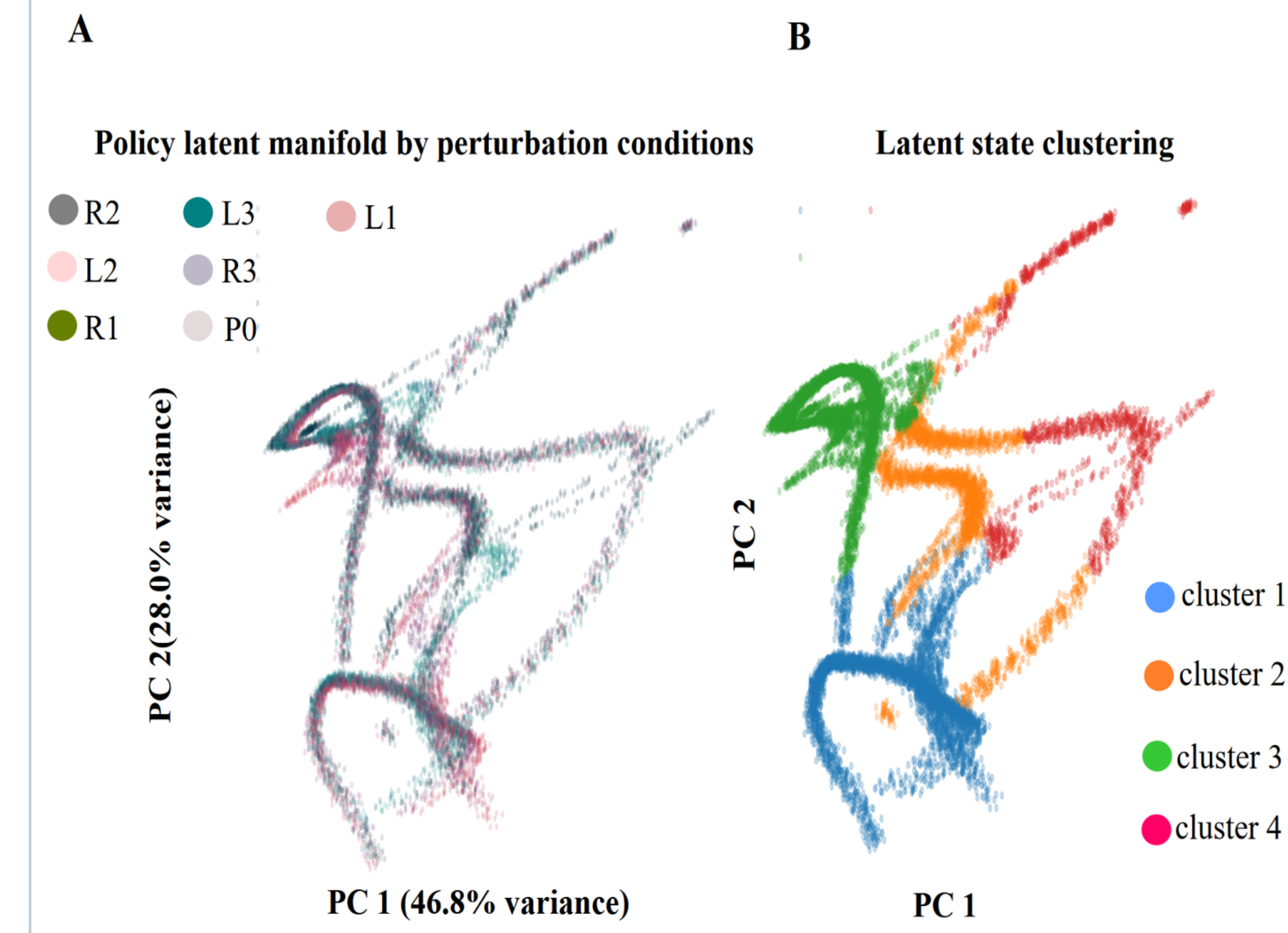
Fig 3 · Velocity Profiles: Proportional Correction, Preserved Speed



(A) Lateral velocity diverges by direction during perturbation window ($t = 50\text{--}55$) · (B) Total speed is invariant — adaptation is purely trajectory-level

RESULTS (cont.)

Fig 4 · Low-Dimensional Manifold (PC1 = 46.8 %, PC2 = 28.0 %)



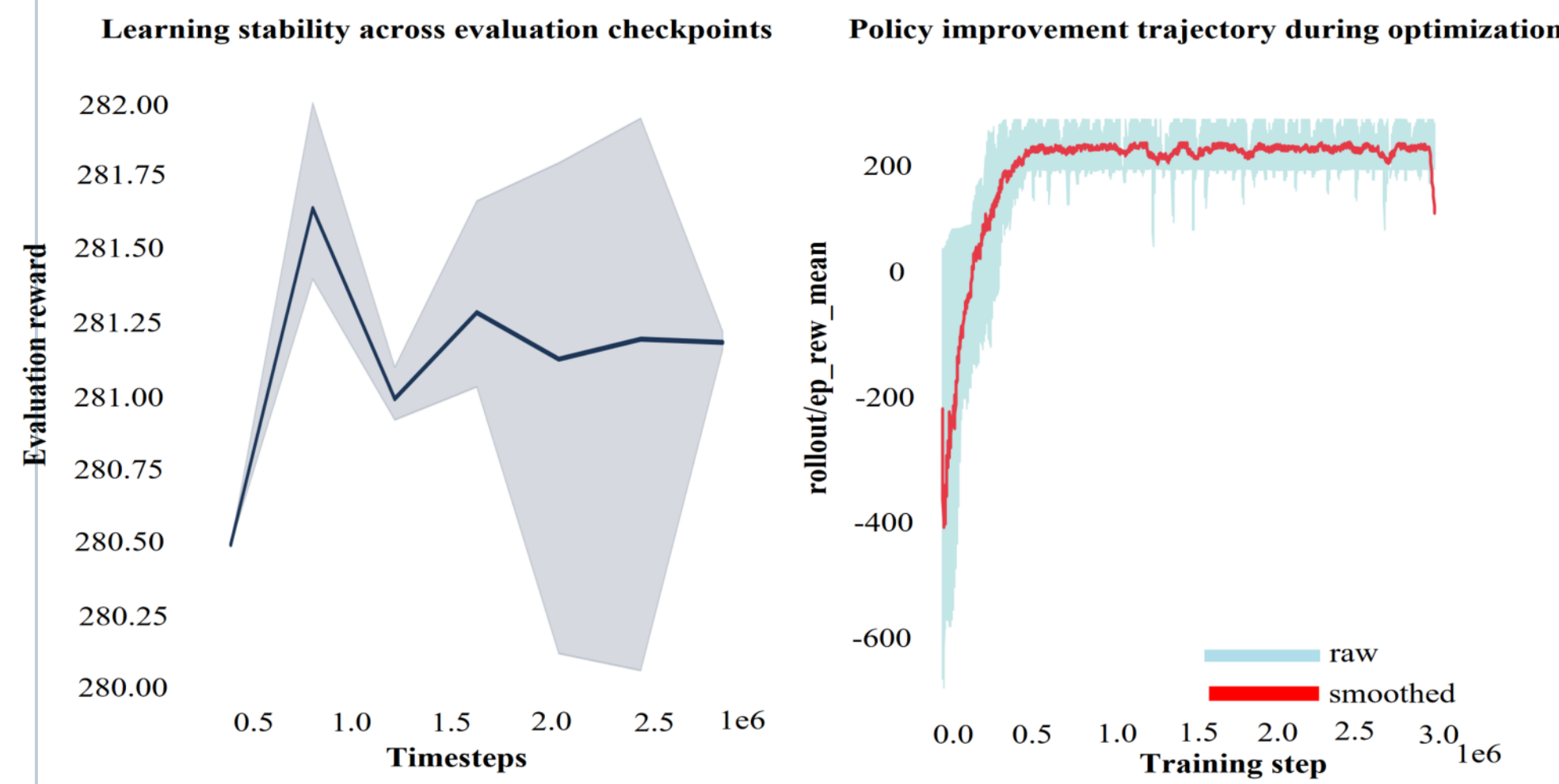
(A) Coloured by condition · (B) K-means $K=4$ reveals phase-aligned clusters

Fig 5 · Linear Decoder Confusion Matrix

		Decoder confusion matrix						
True condition	P0	0.00	0.10	0.03	0.20	0.51	0.00	0.15
	L1	0.00	0.12	0.07	0.35	0.33	0.00	0.12
	L2	0.01	0.08	0.11	0.43	0.30	0.00	0.07
	L3	0.00	0.04	0.04	0.70	0.17	0.00	0.05
	R1	0.01	0.03	0.02	0.22	0.53	0.01	0.18
	R2	0.01	0.04	0.01	0.20	0.50	0.04	0.20
	R3	0.01	0.03	0.02	0.18	0.13	0.01	0.62
		P0	L1	L2	L3	R1	R2	R3

L3 decoded at 70 %, R3 at 62 % · Mild conditions overlap geometrically

Fig 6 · Training Convergence & Evaluation Stability



Rapid convergence in first 1×10^6 steps · Stable plateau validates analyses

DISCUSSION & FUTURE WORK

- Adaptation via trajectory correction with preserved speed mirrors optimal feedback control principles in biological reaching.
- 74.8 % variance in 2 PCs matches motor cortex manifold compression documented in non-human primates and humans.
- Phase-aligned cluster structure parallels sequential neural population states identified during biological reaching.
- Direction-dependent failure asymmetry reflects learned geometry — not an inherent algorithmic limitation.

FUTURE DIRECTIONS

- Multi-seed validation to confirm representational stability.
- Recurrent architectures (LSTM) for temporal perturbation history.
- Human EMG manifold comparison in stroke rehabilitation protocol.
- Closed-loop VR platform using latent geometry as adaptive-difficulty driver.

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Scan for guide & data



Code & data: github.com/OhuePeter/NeuroRL-ObstacleAvoidance-v1.0

Selected references: [1] Churchland et al., Nature 2012 · [2] Gallego et al., Neuron 2017 · [3] Sadleir et al., Nature 2014 · [4] Schulman et al., arXiv 2017 · [5] Pruszyński & Scott, Exp Brain Res 2012 · [6] Oby et al., PNAS 2019 · [7] Todorov & Jordan, Nat Neurosci 2002